A) Unit-тесты / Интеграционные тесты

1. Тестирование Конфигурации (`config`)

- **Файлы:** `test_config.py`
- **Что проверяет:** Правильность загрузки JSON-файлов, валидацию полей (имя, URL, путь к парсеру).
- **Результат:** Все тесты в этом блоке **PASSED**.

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_config.py --cov=ti_framework --cov-report=term-missing -v
============================================================= test session starts ==============================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 6 items

tests/test_config.py::TestLoadSourceConfigs::test_load_valid_json_list PASSED                                                          [ 16%]
tests/test_config.py::TestLoadSourceConfigs::test_load_valid_json_object_with_sources PASSED                                           [ 33%]
tests/test_config.py::TestLoadSourceConfigs::test_load_missing_file_raises_error PASSED                                                [ 50%]
tests/test_config.py::TestSourceConfigModelValidation::test_valid_config_creation PASSED                                               [ 66%]
tests/test_config.py::TestSourceConfigModelValidation::test_config_with_empty_name_raises_error PASSED                                 [ 83%]
tests/test_config.py::TestSourceConfigModelValidation::test_config_with_empty_parser_path_raises_error PASSED                          [100%]
```

2. Тестирование Моделей и Валидации (`domain`)

- **Файлы:** `test_domain.py` (теперь это будет находиться под именем, соответствующим вашему коду)
- **Что проверяет:** Правильность создания объектов `Snapshot`, проверка на валидность данных (например, наличие часового пояса).
- **Результат:** Все тесты в этом блоке **PASSED**.

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_domain.py --cov=ti_framework --cov-report=term-missing -v
============================================================= test session starts ==============================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 3 items

tests/test_domain.py::TestSnapshotModelValidation::test_valid_snapshot_creation PASSED                                                 [ 33%]
tests/test_domain.py::TestSnapshotModelValidation::test_snapshot_with_naive_datetime_raises_error PASSED                               [ 66%]
tests/test_domain.py::TestSnapshotModelValidation::test_snapshot_with_empty_source_name_raises_error PASSED                            [100%]C
:\PyProjects\study\Software-engineering\venv\Lib\site-packages\coverage\inorout.py:577: CoverageWarning: Module ti_framework was previously imported, but not measure
d (module-not-measured); see https://coverage.readthedocs.io/en/7.13.5/messages.html#warning-module-not-measured
  self.warn(msg, slug="module-not-measured")
```

3. Тестирование Инфраструктуры и Хранения (`infrastructure`)

- **Файлы:** `test_infrastructure.py`
- **Что проверяет:** Функциональность файлового хранилища (запись/чтение), работу парсера конкретного типа (`Sec1275Parser`).
- **Результат:** Все тесты в этом блоке **PASSED**.

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_infrastructure.py --cov=ti_framework --cov-report=term-missing -v
============================================================= test session starts ==============================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 3 items

tests/test_infrastructure.py::TestSec1275Parser::test_parse_index_extracts_entries PASSED                                              [ 33%]
tests/test_infrastructure.py::TestFileSystemSnapshotStorage::test_save_and_load_snapshot PASSED                                        [ 66%]
tests/test_infrastructure.py::TestFileSystemSnapshotStorage::test_list_snapshots PASSED                                                [100%]
```

4. Тестирование Портов и Моков (`ports`)

- **Файлы:** `test_ports.py`
- **Что проверяет:** Изолированную логику компонентов, используя моки: как работает HTTP ответ (`HttpResponse`), как парсер
- обрабатывает данные, что делает скраппер при вызовах хранилища и т.д.
- **Результат:** Все тесты в этом блоке **PASSED**.

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_ports.py --cov=ti_framework --cov-report=term-missing -v
============================================================================ test session starts ============================================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 7 items

tests/test_ports.py::test_http_response_success_and_fallback[utf-8] PASSED                                                                                          [ 14%]
tests/test_ports.py::test_http_response_success_and_fallback[None] PASSED                                                                                            [ 28%]
tests/test_ports.py::test_preprocessor_returns_clean_data PASSED                                                                                                     [ 42%]
tests/test_ports.py::test_parser_index_returns_list PASSED                                                                                                           [ 57%]
tests/test_ports.py::test_scrapper_save_snapshot_flow PASSED                                                                                                         [ 71%]
tests/test_ports.py::test_stix_bundle_builder_build_empty_bundle PASSED                                                                                              [ 85%]
tests/test_ports.py::test_stix_bundle_builder_build_with_entries PASSED                                                                                              [100%]
```

5. Тестирование Бизнес-Логики (Пайплайн/Интеграция)

- `test_application.py`: Проверка логики оркестратора (`PipelineRunner`) — как он запускает шаги и управляет состоянием. **PASSED**.

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_application.py --cov=ti_framework --cov-report=term-missing -v
============================================================================ test session starts ============================================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 3 items

tests/test_application.py::TestPipelineRunnerIntegration::test_run_source_creates_snapshot_and_parses_index PASSED                                                   [ 33%]
tests/test_application.py::TestPipelineRunnerIntegration::test_run_source_drops_fresh_snapshot_if_no_new_entries PASSED                                              [ 66%]
tests/test_application.py::TestPipelineRunnerIntegration::test_run_all_filters_disabled_sources PASSED                                                               [100%]C
:\PyProjects\study\Software-engineering\venv\Lib\site-packages\coverage\inorout.py:577: CoverageWarning: Module ti_framework was previously imported, but not measure
d (module-not-measured); see https://coverage.readthedocs.io/en/7.13.5/messages.html#warning-module-not-measured
  self.warn(msg, slug="module-not-measured")
```

- `test_resilience.py`: Проверка полного цикла работы системы: как пайплайн управляет сбоями и агрегирует данные. **PASSED**.

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_pipeline_resilience.py --cov=ti_framework --cov-report=term-missing -v
============================================================================ test session starts ============================================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 1 item

tests/test_pipeline_resilience.py::PipelineResilienceTest::test_failed_source_does_not_stop_other_sources PASSED                                                     [100%]
```

- `test_ioc_filter.py`: Проверка, что фильтры (IP/URL/Domain) работают корректно в пайплайне. **PASSED**. **(Интеграционный тест)**

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_ioc_filter.py --cov=ti_framework --cov-report=term-missing -v
================================================================= test session starts =================================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 5 items

tests/test_ioc_filter.py::test_drop_special_purpose_ipv4_rule_blocks_ioc_values_from_reserved_ranges PASSED                                    [ 20%]
tests/test_ioc_filter.py::test_drop_special_purpose_ipv4_rule_keeps_public_ipv4_and_explicit_iana_exceptions PASSED                            [ 40%]
tests/test_ioc_filter.py::test_drop_invalid_domain_rule_blocks_malformed_and_internal_domains PASSED                                          [ 60%]
tests/test_ioc_filter.py::test_drop_internal_url_rule_blocks_local_invalid_and_private_url_hosts PASSED                                       [ 80%]
tests/test_ioc_filter.py::test_rule_based_ioc_filter_filters_entry_without_touching_other_fields PASSED                                        [100%]
```

- `test_ioc_filter_pipeline_integration.py`: Проверка, что фильтры (IP/URL/Domain) работают корректно в пайплайне. **PASSED**.

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_ioc_filter_pipeline_integration.py --cov=ti_framework --cov-report=term-missing -v
================================================================= test session starts =================================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 1 item

tests/test_ioc_filter_pipeline_integration.py::test_pipeline_applies_all_ioc_quality_filters_before_stix_bundle_build PASSED                  [100%]
```

- `test_e2e_stix_pipeline.py`: Полный сквозной тест с моками HTTP-клиента для проверки полного цикла:
  Скрапер ⟶⟶ Парсер ⟶⟶ Фильтр ⟶⟶ STIX Bundle. **PASSED**.
  (End-to-End тест)

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_e2e_stix_pipeline.py --cov=ti_framework --cov-report=term-missing -v
================================================================= test session starts =================================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 1 item

tests/test_e2e_stix_pipeline.py::StixPipelineE2ETest::test_pipeline_builds_stix_bundle_on_first_run_and_skips_second_run PASSED                [100%]
```

6. Тестирование CLI и Утилиты

- **Файлы:** `test_cli_run.py`, `test_cli_status.py`, `test_cli_validate.py`
- **Что проверяет:** Правильность работы командной строки (CLI) — корректный выход кода, правильный вывод статусов в файл и консоль. **PASSED**.

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_cli_run.py --cov=ti_framework --cov-report=term-missing -v
================================================================= test session starts =================================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 3 items

tests/test_cli_run.py::test_run_command_prints_successful_result PASSED                                                                       [ 33%]
tests/test_cli_run.py::test_run_command_returns_nonzero_when_any_source_failed PASSED                                                          [ 66%]
tests/test_cli_run.py::test_run_command_aborts_on_invalid_config PASSED                                                                        [100%]
```

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_cli_status.py --cov=ti_framework --cov-report=term-missing -v
============================================================= test session starts ==============================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 2 items

tests/test_cli_status.py::test_run_command_writes_status_file_and_status_reads_it PASSED                                                [ 50%]
tests/test_cli_status.py::test_status_command_returns_nonzero_for_missing_status_file PASSED                                            [100%]
```

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_cli_validate.py --cov=ti_framework --cov-report=term-missing -v
============================================================= test session starts ==============================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 4 items

tests/test_cli_validate.py::test_validate_config_accepts_valid_config PASSED                                                            [ 25%]
tests/test_cli_validate.py::test_validate_command_reports_invalid_parser PASSED                                                         [ 50%]
tests/test_cli_validate.py::test_validate_command_reports_invalid_source_url PASSED                                                     [ 75%]
tests/test_cli_validate.py::test_validate_command_returns_zero_for_valid_config PASSED                                                  [100%]
```

7. Тестирование конфигурации логгера

- **Файлы:** `test_logging_configuration.py`
- **Что проверяет:** Проверка работоспособности логгера. **PASSED**.

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_logging_configuration.py --cov=ti_framework --cov-report=term-missing -v
============================================================= test session starts ==============================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 5 items

tests/test_logging_configuration.py::test_normalize_log_level_accepts_case_insensitive_names PASSED                                    [ 20%]
tests/test_logging_configuration.py::test_configure_framework_logging_attaches_handler PASSED                                          [ 40%]
tests/test_logging_configuration.py::test_configure_framework_logging_uses_compact_format_for_info PASSED                              [ 60%]
tests/test_logging_configuration.py::test_configure_framework_logging_uses_verbose_format_for_debug PASSED                             [ 80%]
tests/test_logging_configuration.py::test_configure_framework_logging_reconfigures_existing_handler PASSED                              [100%]
```

8. Парсинг данных (Parser)

- **Файлы:** `test_securelist_parser.py` / `test_proofpoint_threat_insight_parser.py`
- **Что проверяет:** Способность извлечь структурированные данные (IOCs) как из списка тегов, так и из табличных форматов, из форматов HTML/XML. **PASSED**.

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_securelist_parser.py --cov=ti_framework --cov-report=term-missing -v
============================================================= test session starts ==============================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 2 items

tests/test_securelist_parser.py::SecurelistParserTest::test_parse_entry_extracts_basic_iocs PASSED                                      [ 50%]
tests/test_securelist_parser.py::SecurelistParserTest::test_parse_index_extracts_publication_links PASSED                               [100%]
```

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_proofpoint_threat_insight_parser.py --cov=ti_framework --cov-report=term-missing -v
============================================================= test session starts ==============================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 4 items

tests/test_proofpoint_threat_insight_parser.py::ProofpointThreatInsightParserTest::test_parse_entry_extracts_iocs_from_example_section_text PASSED [ 25%]
tests/test_proofpoint_threat_insight_parser.py::ProofpointThreatInsightParserTest::test_parse_entry_extracts_iocs_from_indicator_table_without_heading PASSED [ 50%]
tests/test_proofpoint_threat_insight_parser.py::ProofpointThreatInsightParserTest::test_parse_entry_extracts_iocs_from_table PASSED [ 75%]
tests/test_proofpoint_threat_insight_parser.py::ProofpointThreatInsightParserTest::test_parse_index_extracts_publication_links PASSED [100%]
```

Б) Smoke-тест

- **Файлы:** `test_smoke.py`
- **Что проверяет:** Загрузка основных модулей: config, domain, application. **PASSED**.

```
(venv) PS C:\PyProjects\study\Software-engineering> python -m pytest tests/test_smoke.py --cov=ti_framework --cov-report=term-missing -v
============================================================= test session starts ==============================================================
platform win32 -- Python 3.12.8, pytest-9.0.2, pluggy-1.6.0 -- C:\PyProjects\study\Software-engineering\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\PyProjects\study\Software-engineering
configfile: pyproject.toml
plugins: cov-7.1.0
collected 2 items

tests/test_smoke.py::test_app_imports PASSED                                                                                          [ 50%]
tests/test_smoke.py::test_pipeline_initialization PASSED                                                                              [100%]
```